Spatial Analysis |

Proximity – Overlay – Extract

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→ DATA LINK

01

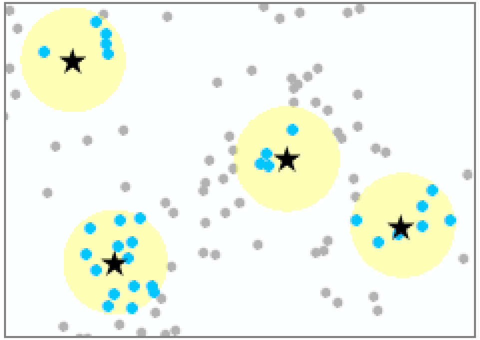
Proximity Analysis

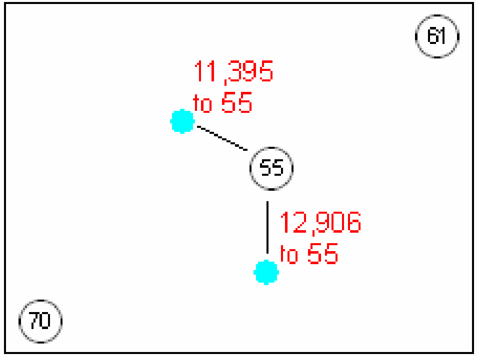
Proximity Analysis

One of the most basic questions asked of a GIS is
"What's near what?"

That's exactly what **proximity analysis** answers
It helps determine things like

- What is the distance between two locations?
- What is the nearest or farthest feature from something?





Proximity Tools | Feature-based

01 Select by Location

02 Buffer | Graphic Buffer | Multiple Ring Buffer

03 Thiessen Polygons (Voronoi)

04 Generate Origin-Destination Links

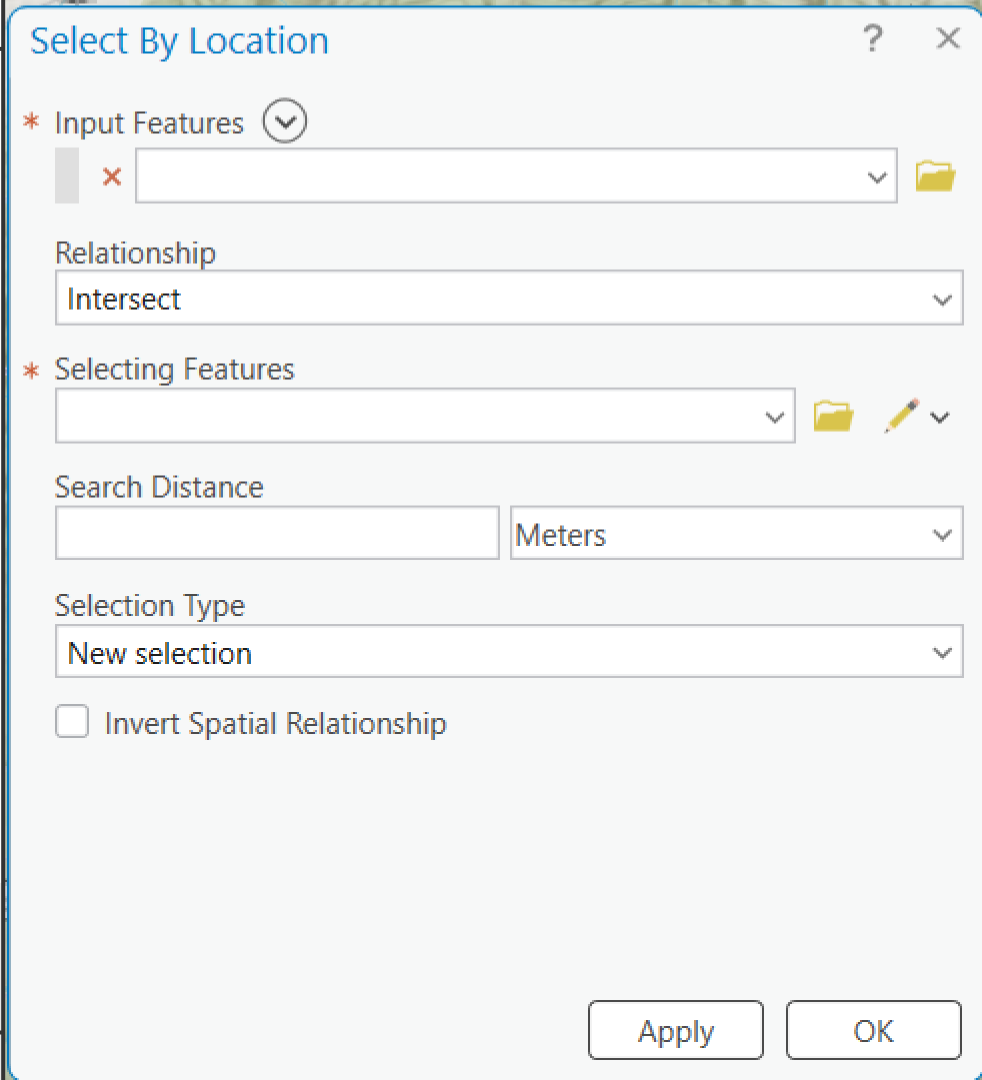
05 Near | Generate Near Table

06 Polygon Neighbors

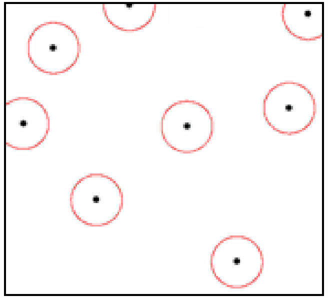
01

Select by Location

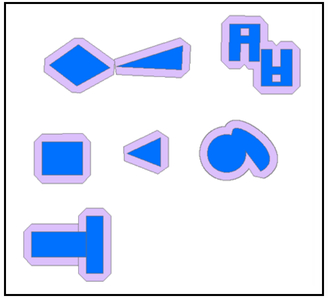
The Select Layer By Location tool lets you select features in one layer based on their spatial relationship
(e.g., intersect, contain, within distance)
with features in another layer.
It’s used for queries like:
“Which homes are within the flood boundary?”
or “Which roads intersect forest areas ?”



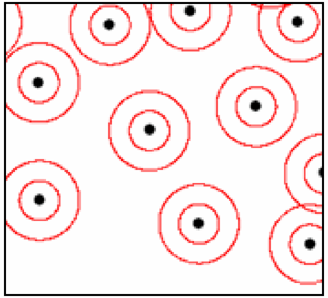
02 Buffer | Graphic Buffer | Multiple Ring Buffer



Buffer : Creates a polygon around input features (points, lines, or polygons) at a specified distance. Useful for identifying what's nearby .
e.g., all houses within 500 meters of a hospital



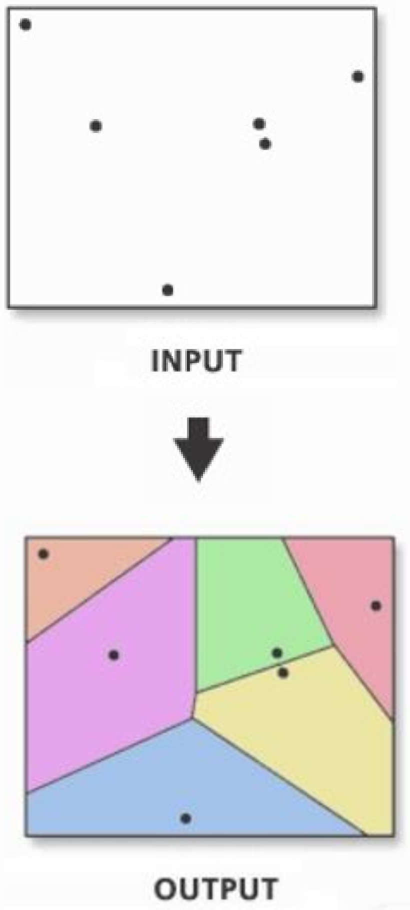
Graphic Buffer : Like Buffer but focuses on customizing the visual appearance of the buffer for better map presentation.



Multiple Ring Buffer : Generates several buffer zones at different distances (e.g., 100m, 250m, 500m) to represent gradual levels of spatial influence

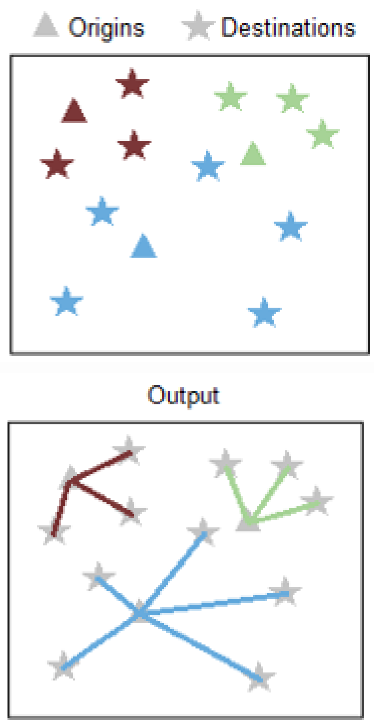
03 Thiessen Polygons (Voronoi)

Thiessen Polygons divides space into polygons based on a set of points,
where each polygon contains the area closest to a specific point than to any other. Useful for defining zones of influence or service areas, such as the nearest fire station or hospital for each location



04 Generate Origin-Destination Links

A tool that draws lines between origin points and destination points to represent direct connections.
Used for movement or accessibility analysis, such as linking homes to the nearest bus stops or farms to the nearest markets.
Can be based on straight-line distance or a road network

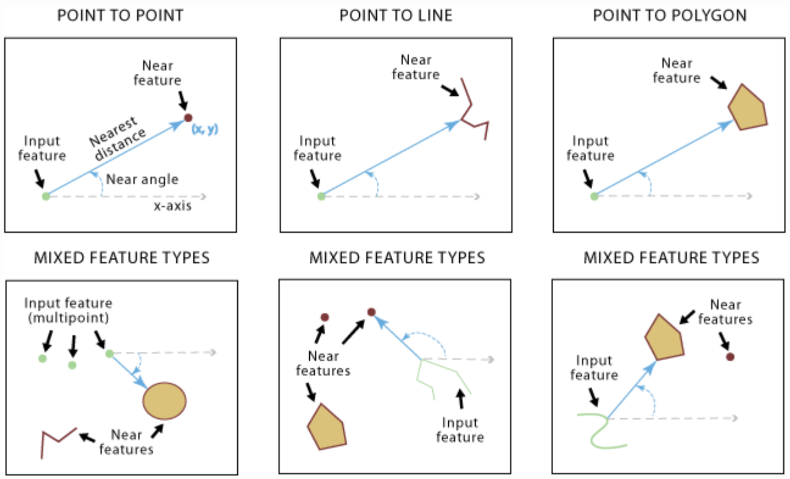


05 Near | Generate Near Table

Near : Calculates the closest feature from one or more layers and stores the distance and the closest feature's ID directly in the input feature's attribute table

Generate Near Table : Creates Standalone Table listing distances and nearest features for each input, with the option to include multiple nearest features per record

(Numeric Value Output)



wells_500m_off_Roads

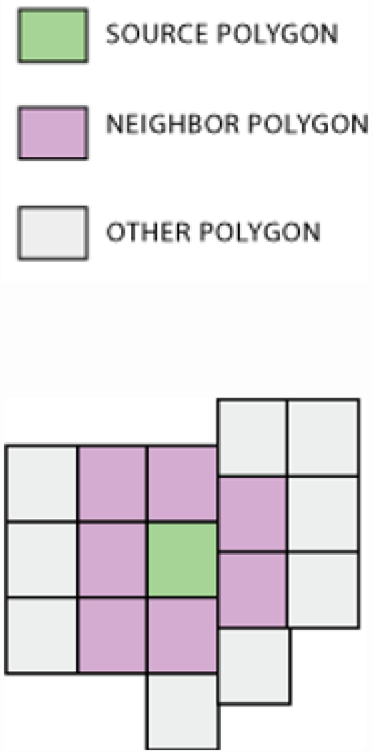
Field: Add Delete Calculate Selection: Zoom To Switch Clear Delete

OBJECTID	IN_FID	NEAR_FID	NEAR_DIST	FROM_X	FROM_Y	NEAR_X	NEAR_Y	NEAR_ANGLE
4	4	16	365.520914	28596.578551	43905.709119	28938.339679	44035.342718	20.772265
5	5	18	200.689083	29562.937511	43882.139284	29362.594736	43870.354531	-176.633572
6	6	21	163.72049	29775.065039	42703.652796	29633.646687	42786.146725	149.743596
7	7	28	310.011427	29327.240148	41866.927107	29633.646687	41819.787766	-8.746139
8	8	24	295.328005	28066.259402	42350.106915	28325.5266	42208.688563	-28.610439
9	9	35	374.528985	27276.673383	42974.704747	27359.167312	43340.035709	77.275668
10	10	10	251.931891	27877.701708	44388.888599	27630.219263	44341.749257	-169.215741
11	11	1	200.342447	27971.980391	43893.924365	27771.637944	43893.924365	180
12	12	13	328.711357	28254.817424	43398.959804	27936.625803	43316.465875	-165.465578
13	13	2	285.037802	28549.438882	42880.425736	28584.79347	43163.262441	82.874984

06 Polygon Neighbors

Generates a table that lists each polygon along with all polygons that share a boundary (edge) or a vertex with it. The output includes details like the neighbor polygon ID, the length of shared borders, and the type of adjacency (edge or corner)

(Numeric Value Output)



Challenge

What tool would you use to create 300-meter coverage areas around each proposed bus station location ?

Challenge

Which cities are located within a 1-kilometer distance of the rivers?
What is the most suitable tool to answer this question

Challenge

How can you define zones of influence for each hospital, where each zone represents the area closest to that specific hospital ?

Task 1

Gym Expansion - Proximity Analysis

A fitness gym in Wilmington, North Carolina, wants to expand its locations.
As a GIS Analyst, you have been asked to collect data supporting the placement of the new location based on coverage (5 sm) gaps. You will use proximity analysis to answer the following

- Which gym location is the closest to each member's residence?
- Which areas have gym location coverage gaps?
- Which members do not live within the gym location coverage areas?

02

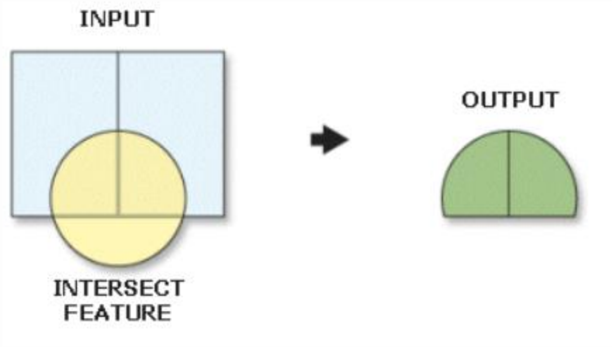
Overlay Analysis

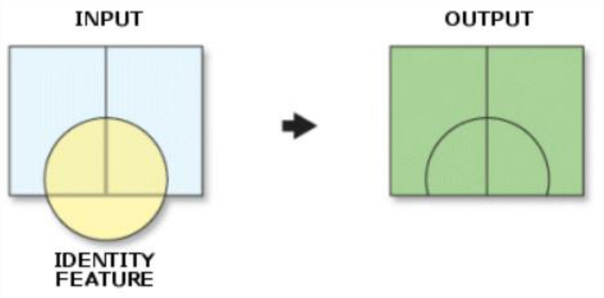
Overlay Analysis

" What's on top of what ? "

That's exactly what **overlay analysis** answers.
It helps determine things like

- Where features overlap?
- Where features are within another feature?
- Where features are outside the extent of another feature?





Overlay Analysis Tools

01 Erase | Pairwise Erase

02 Intersect | Pairwise Intersect

03 Symmetrical Difference

04 Union

05 Update

06 Identity

07 Remove Overlap (multiple)

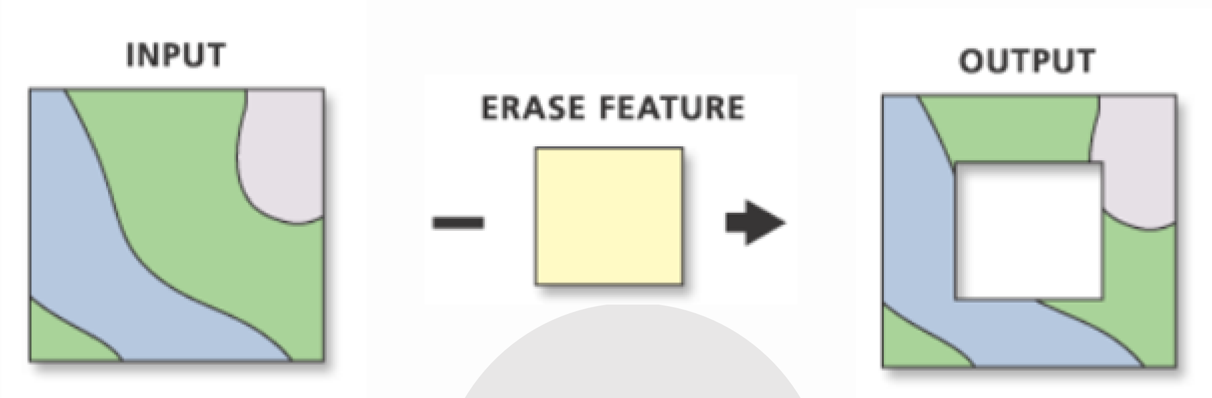
08 Spatial Join

01

Erase | Pairwise Erase

Erase : Removes areas of the input layer that overlap with the erase layer, keeping only the non-overlapping portions. Useful for excluding certain regions from analysis

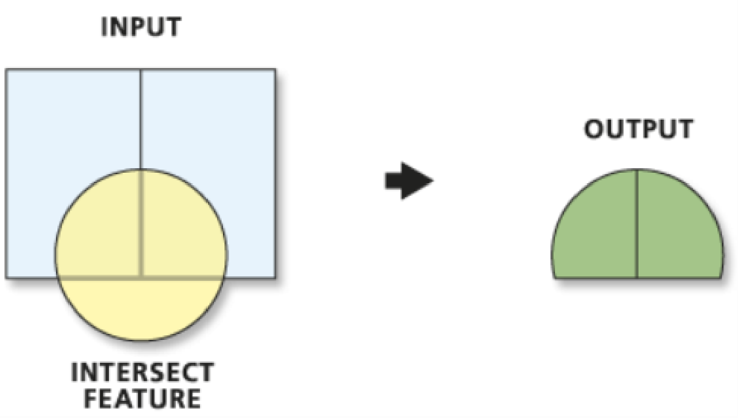
Pairwise Erase : Performs the same function as Erase but optimized for large datasets by processing features in pairs, which can improve performance.



02 Intersect | Pairwise Intersect

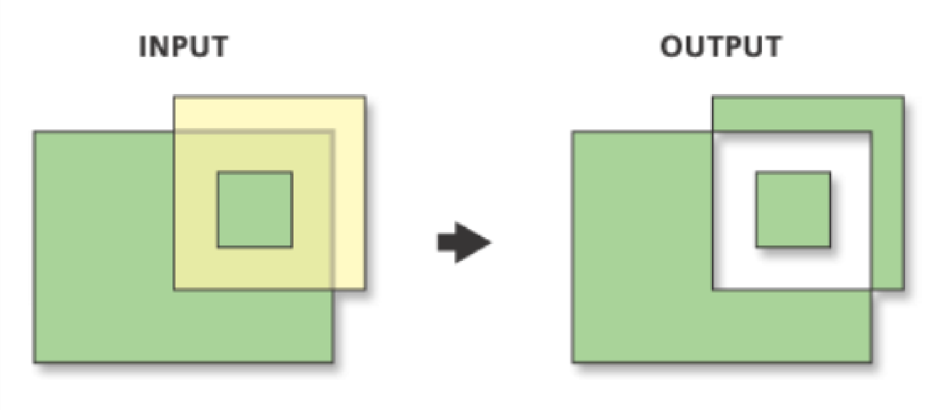
Intersect : Generates features where input layers overlap, keeping only the common areas and attributes from all layers

Pairwise Intersect : Same as Intersect but designed for faster processing on large datasets by handling features in pairs



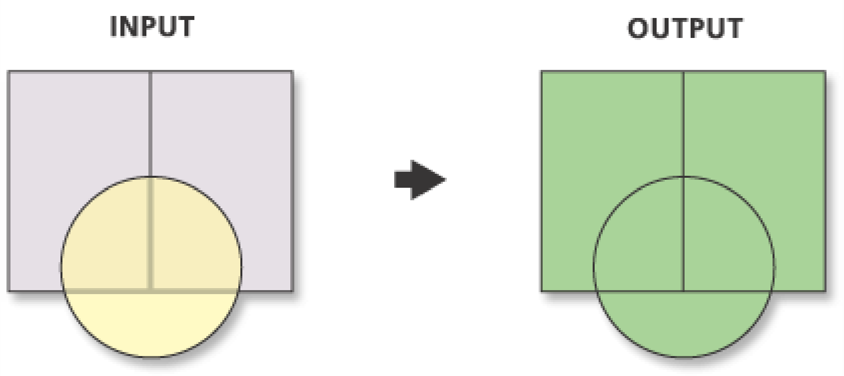
03 Symmetrical Difference

Creates features from areas that do not overlap between input layers, keeping only the non-shared parts



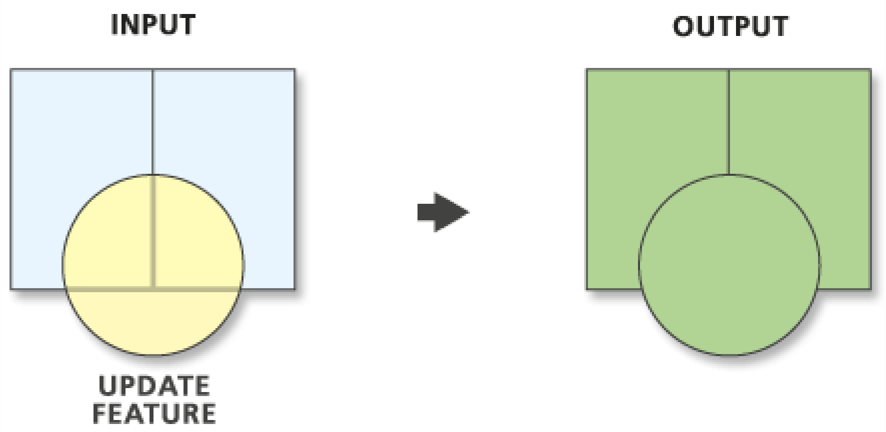
04 Union

Combines all areas from the input layers into a single layer, keeping both overlapping and non-overlapping areas with attributes from all layers



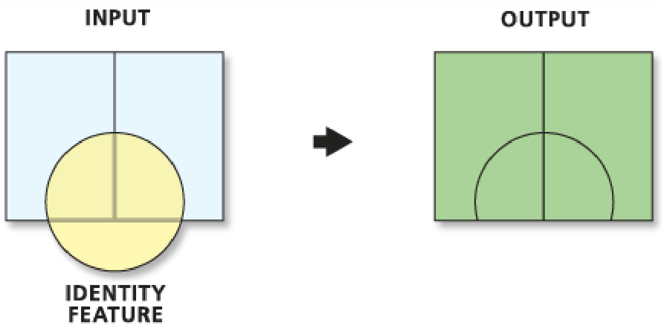
05 Update

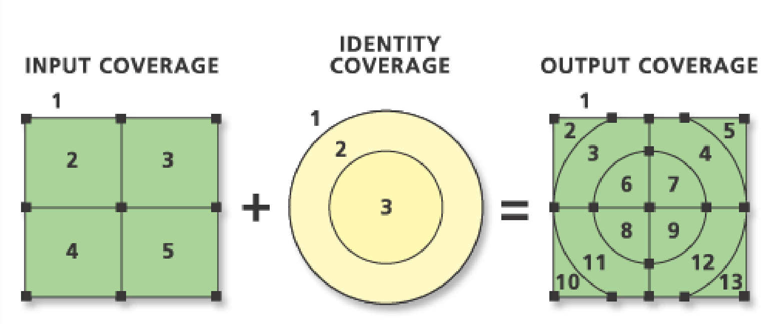
Replaces portions of the input layer with features from the update layer where they overlap.



06 Identity

Overlays the input layer with an identity layer and keeps all input features, adding attributes from the identity layer where they overlap

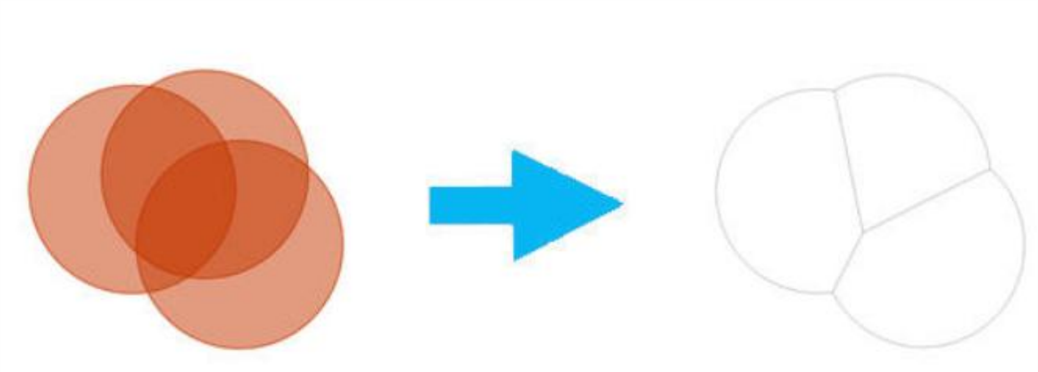




07 Remove Overlap (multiple)

Eliminates overlapping areas between multiple layers, ensuring features don't share the same space.

The layer must not contain a Z value.



08 Spatial Join

Transfers attributes from one layer to another based on spatial relationships, such as intersection or proximity

Challenge

What tool removes overlapping parts of a layer based on another layer ?

Challenge

What tool combines all features from two layers into a single layer while retaining attributes from both ?

Challenge

What is the appropriate tool to use to identify the areas where agricultural lands and forests overlap ?

Task 2

Ambulance Site Selection – Overlay Analysis

The vice minister of health in Loudoun County needs to identify a location for a second Ambulance unit within the county.
The site must meet the following requirements:

- **Be within one mile of main roads**
- **Be more than a 20-minute car ride from the current main hospital**
- **Be within urban areas**

04

Extract Analysis

Extract Analysis Tools

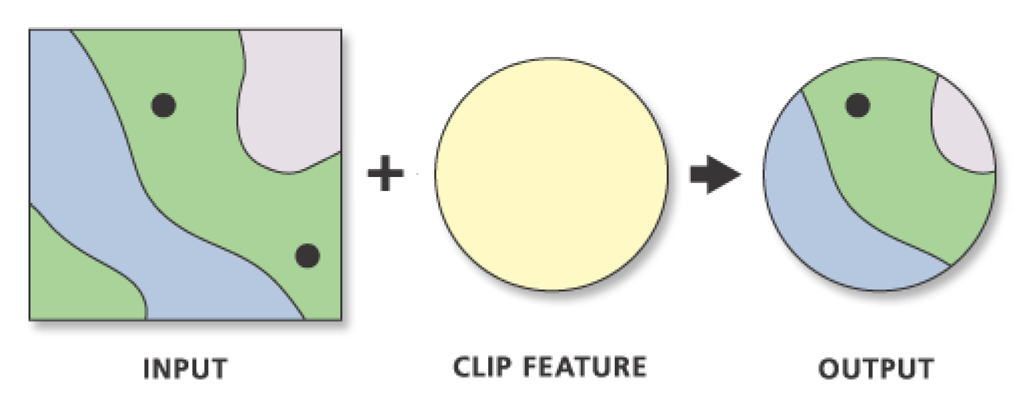
01 Clip

02 Split By Attributes

03 Table Select

01 Clip | Pairwise Clip

Extracts features from one layer that fall within the boundary of another layer, keeping only the overlapping parts. This is useful for focusing on a specific area of interest



02 Split By Attributes

Divides a dataset into multiple smaller datasets based on a specified attribute field. Each output dataset contains features sharing the same attribute value

03 Table Select

Selects records from a table or attribute table based on a SQL expression, creating a new table with only the matching rows

Challenge

How can you create a new table containing only the records of schools with more than 500 students ?

Challenge

How can you extract all the features that fall within the boundaries of a specific administrative area ?

Challenge

You have a point layer representing homes and a polygon layer representing administrative districts. You want to add the administrative district name to each home's attribute table. Which tool is the most appropriate for this?

Task 3

Flood-Affected Cities

Using what you have learned in previous lessons, answer the following question:
Which cities are likely to be affected if a 7 km flood occurs along any of the rivers?

Lab 1

Real Estate Analysis Lab

A real estate investor is looking at property in Oregon
You have been hired to collect critical assessment data for several parcels that are currently for sale.
Using what you learned in previous lessons, answer the following

- **How many acres of the parcels for sale are outside the floodplain (the usable area)?**
  Hint: Use the Make Feature Layer tool and apply a Ratio policy to the GIS_ACRES field.
- **How many acres of the parcels are inside the floodplain (the flooded area)?**
- **Which buildings are within the floodplain, and which buildings are outside the floodplain?**

Form Lab

Using Features layers in form_data dataset answer the following questions in this form

LINK

Thanks!

Do you have any questions?

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